Kappa-Framework for Higgs Measurements

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Kappa- framework

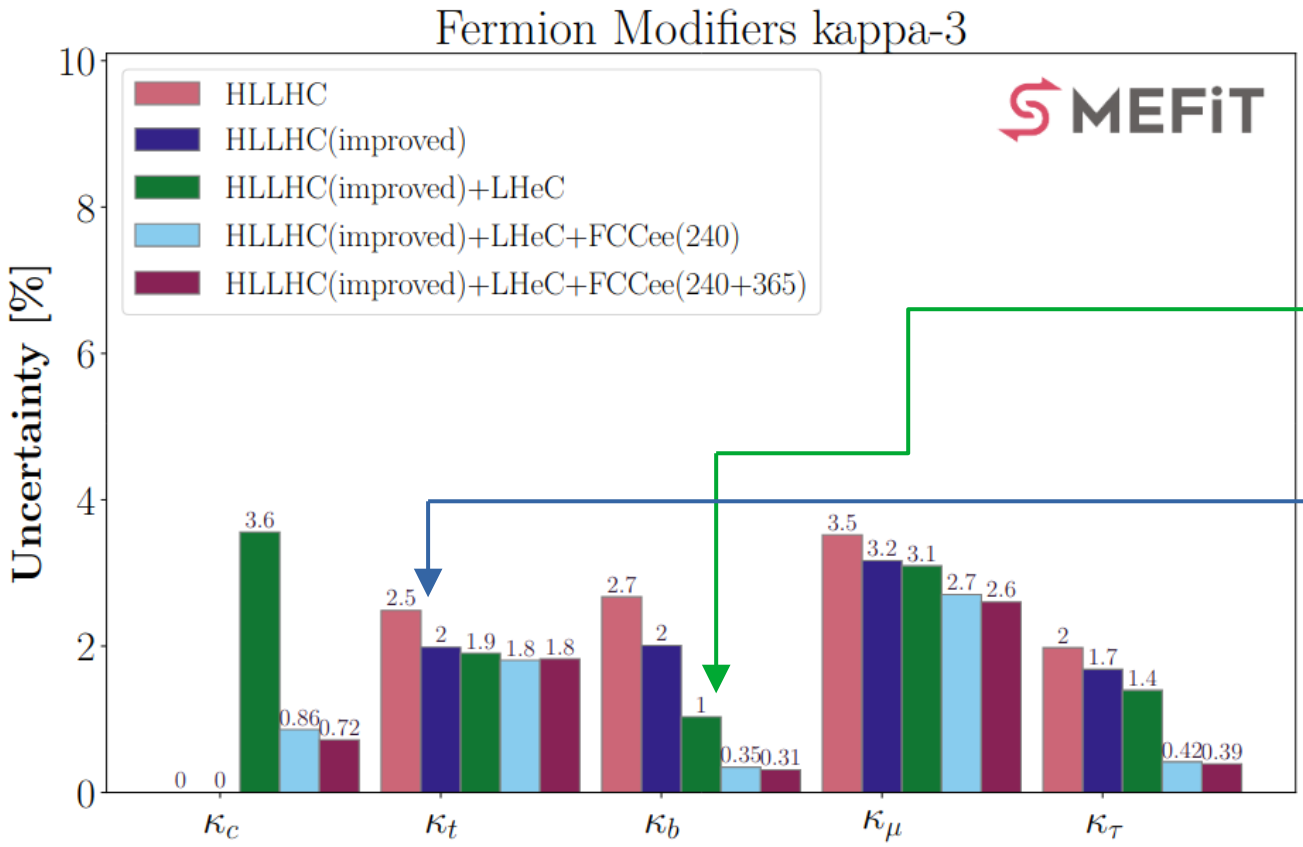
- Rescale production mode
- Rescale widths

$$\mu_i^f = \frac{\kappa_i^2 \kappa_f^2}{\kappa_H^2}$$

	Observable	LHeC	FCC-eh
$\kappa_i=\kappa_W$	$\delta\mu_{WBF,bb}$	0.008	0.0025
	$\delta\mu_{WBF,cc}$	0.071	0.022
	$\delta\mu_{WBF,gg}$	0.058	0.018
$\kappa_i=\kappa_Z$	$\delta\mu_{ZBF,bb}$	0.023	0.0065
	$\delta\mu_{WBF,WW}$	0.062	0.019
	$\delta\mu_{WBF,ZZ}$	0.120	0.038
	$\delta\mu_{WBF,\tau\tau}$	0.052	0.016
	$\delta\mu_{WBF,\gamma\gamma}$	0.15	0.046
	$\delta\mu_{ZBF,cc}$	0.200	0.058
	$\delta\mu_{ZBF,gg}$	0.160	0.047
	$\delta\mu_{ZBF,WW}$	0.170	0.050
	$\delta\mu_{ZBF,ZZ}$	0.350	0.100
	$\delta\mu_{ZBF,\tau\tau}$	0.15	0.042
	$\delta\mu_{ZBF,\gamma\gamma}$	0.42	0.120

[ESPPU2020]

LheC and Higgs coupling

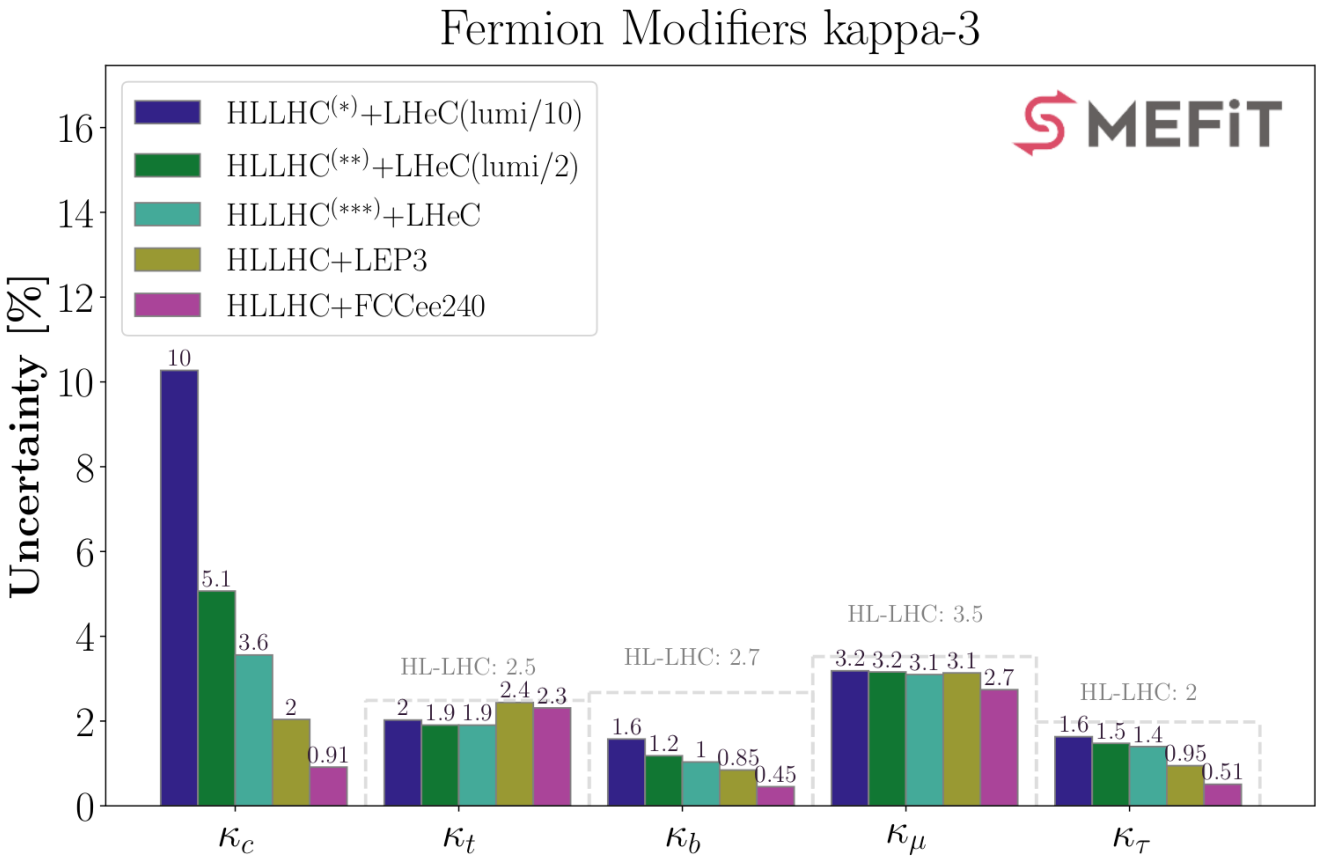


- Allow precise measurements of Higgs coupling
- α_s and PDF measurements reduce HL uncertainties

Ingredients needed and prospect

- Signal-strength input similar to the LHeC one for ESPPU2020

- After that we can see how lumi affects:
    - HL uncertainties reduction
    - LhuC H couplings measurements



Ingredients needed and prospect

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What we have already:

- Fitting Framework (SMEFiT)
- HL-LHC projections
- HL errors breakdown

Given different lumi scenarios we can immediately see the improvements in HL even without LhuC projections

CAUTION: LhuC itself will give the leading constraint for most couplings, not HL-LHC.